Where to Fly the Cell Tower: Terrain-Dependent Positioning of a Nomadic 5G Node Serving Ground Robots in Disaster Search-and-Rescue

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***Abstract*—When an earthquake, flood, or fire destroys or overwhelms terrestrial cellular infrastructure, the ground robots increasingly used for search and rescue lose the very connectivity they depend on to stream sensor data out and receive commands in. A base station carried by an uncrewed aerial vehicle, a nomadic node, can restore service on demand, but because the node is itself mobile a responder must decide where to fly it, and that decision interacts with the terrain the robots operate in. This paper studies that decision empirically. We build a protocol-in-the-loop emulation that couples a physics simulator to a real fifth-generation standalone network stack: the three-dimensional motion of an aerial base station and two ground-robot devices continuously drives a terrain-parameterised air-to-ground path-loss model whose output is injected at runtime into a live radio access network and core, over which genuine bulk-upload, video, and command traffic are measured in both directions across the emulated air interface. We instantiate five terrains, open field, urban, forest, hills, and an enterable building, each sharing one obstacle geometry between the rendered world and the radio channel, and run each as a six-hour trial split into a static-node phase and a roaming-node phase while both robots patrol. We find that the effect of node mobility on connectivity is strongly terrain-dependent and changes sign with how pervasively the terrain blocks the link: roaming rescues a weak device in line of sight, is neutral under localised building shadowing, and degrades connectivity as blockage becomes pervasive, while deep building penetration is effectively uncoverable at this link budget. We re-analyse the logged data to derive four operational metrics for responders, and we release the testbed, the automation, and the dataset.**

***Index Terms*—UAV-enabled networks, non-terrestrial networks, nomadic base station, aerial base station, 5G NR, air-to-ground channel, propagation, disaster response, search and rescue, protocol-in-the-loop emulation, OpenAirInterface.**

I. Introduction

A. The Disaster-Connectivity Problem

Modern disaster response is increasingly robotic. Ground robots enter collapsed buildings, rubble fields, flood zones, and wildfire margins that are too dangerous for responders, streaming video and sensor data to a command post and receiving navigation waypoints and safety-critical commands in return. This mission is built on an implicit assumption: that a network is there to carry the traffic. In exactly the scenarios where robots are most valuable, that assumption fails. Earthquakes and floods topple or submerge cell sites; fires burn fibre backhaul; a surge of civilian traffic congests whatever survives. The connectivity a rescue team depends on is cut precisely when it is needed most, and a robot that cannot phone home is blind to its operator and deaf to a stop command.

B. UAV-Mounted Nomadic Nodes as a Response

A base station carried aloft by an uncrewed aerial vehicle (UAV), a *nomadic node*, is a direct answer: it can be flown to the incident and provide an on-demand aerial cell over the search area [1], [5], [6]. Its appeal for public safety is well argued—rapid deployment, no dependence on surviving ground infrastructure, and a vantage point that tends toward line of sight [3]. The concept and its analytic air-to-ground (A2G) channel models are mature [4], [7].

C. The Positioning Question

Because the node is airborne and *mobile*, a mission planner faces a decision a fixed tower never poses: as robots spread through the search area, should the node *hover* at a vantage point or *roam* to follow them, and at what altitude and horizontal offset? The naive intuition—follow the robots—is not obviously right, because the robots operate *in terrain*. Buildings cast radio shadows; tree canopies attenuate in proportion to the foliage path length; hills diffract; walls impose penetration loss. Each mechanism loads the A2G channel differently, and moving the node changes the *propagation path* through that terrain, sometimes clearing an obstacle and sometimes introducing one. Whether mobility helps or hurts

should therefore depend on the terrain, and the size and even the sign of that effect is what a responder needs to know before deciding where to fly.

D. Why Simulation, and What It Validates

Answering this with fidelity is difficult. Closed-form analysis and packet-level simulation abstract away the very mechanisms whose behaviour we want to observe: the scheduler's reaction to changing signal quality, hybrid-ARQ retransmission under fading, modulation-and-coding adaptation, and the radio-link-failure (RLF) and re-establishment state machine. A live field trial with a real UAV, across five controlled terrains for many hours each, is expensive, weather-bound, and hard to repeat. We take a middle path: a *protocol-in-the-loop* emulation that runs an unmodified fifth-generation (5G) standalone stack and drives it with physics-simulated robot and UAV motion through a terrain-consistent A2G channel. This validates the *mechanism and the operational rules*—how a complete protocol stack reports connectivity as geometry and terrain change—rather than absolute capacity, and it does so reproducibly.

E. Contributions

This paper makes four contributions.

- A mobility- and terrain-coupled emulation architecture (Sections IV–VI): the full software stack, its inter-component communication, the geometry-to-signal pipeline, and a terrain-parameterised A2G channel whose non-line-of-sight (NLoS) excess is derived and evaluated on the same obstacle geometry the physics world renders.
- A controlled hover-versus-roam study across five terrains (Section XI), each a six-hour trial reporting both devices' throughput, latency, coverage, and outage, establishing that the sign of the mobility effect flips with the pervasiveness of terrain blockage.
- Four operational metrics for disaster response, re-analysed from the logged data (Section XI J): time-to-first-outage, a minimum-viable-position rule, a governor-mediated recovery time (reported with an explicit granularity caveat), and a per-terrain hover-versus-roam decision rule.
- A propagation-aware coverage governor (Section VIII) that keeps robots connected in NLoS by reacting to *measured* path loss, positioned as the rule-based baseline for a future learned controller, together with the emulation correctness safeguards that make single-host measurement honest.

By instantiating five terrains under one instrument, we ask whether a single positioning rule can serve a disaster responder, and find that it cannot: the right choice of where to fly the node depends on the terrain.

II. Related Work

A. UAV-Enabled Networks and A2G Channel Models

Al-Hourani *et al.* [4] derived the widely used elevation-angle model in which the probability of a line-of-sight link to a low-altitude platform is a sigmoid of the elevation angle, and showed that an optimal altitude balances free-space loss against the excess loss of shallow-angle links; our channel core adopts this formulation. The tutorial of Zeng *et al.* [1] and its 5G-and-beyond follow-up [2], and the survey of Mozaffari *et al.* [3], catalogue UAV placement, trajectory, and energy trade-offs, but are predominantly analytic or system-level. 3GPP characterised aerial channels and interference for cellular UAV support [7] and specified the general A2G and outdoor-to-indoor channel families [8].

B. Terrain-Specific Propagation Models

Terrain excess loss has long-standing empirical models we reuse verbatim. Weissberger's modified exponential-decay model [9] gives foliage attenuation as a function of the depth of canopy a ray traverses; the ITU-R knife-edge recommendation [10] gives single-obstacle diffraction loss as a function of the Fresnel-Kirchhoff parameter; and 3GPP TR 38.901 [8] specifies outdoor-to-indoor penetration as a wall term plus a depth-dependent term. These are normally used in isolation; a contribution here is coupling all four to one live stack under a common testbed.

C. UAV-Assisted Disaster and Public-Safety Communications

Naqvi *et al.* [5] argue drone-aided communication as a key enabler for resilient public-safety networks, restoring connectivity in disaster regions and offloading congested cells. Merwaday and Guvenc [6] quantify the throughput and coverage gains of adding aerial base stations when terrestrial infrastructure is damaged. This body of work motivates the deployment but evaluates it under statistical channels; we complement it with protocol-stack measurement across distinct terrains.

D. Multi-Robot Networking for Search and Rescue

On the robotics side, Queralta *et al.* [11] survey collaborative multi-robot search and rescue and note communication as a recurring bottleneck, while connectivity-aware coordination work [11] keeps robot teams within range of one another or a base. That literature typically assumes a disc- or exponential-range communication model; we instead measure, end-to-end on a real stack, how the *base itself* should move given roaming robots, with a channel shaped by real terrain rather than a range disc.

E. Gap This Paper Addresses

To our knowledge, the specific combination—physics-simulated multi-robot mobility continuously modulating a live 5G standalone stack that carries measured bidirectional application traffic, parameterised across five terrains, and framed as a hover-versus-roam positioning question for disaster response—has not previously been characterised. We fill that gap and translate the result into operational rules.

III. The Testbed Platform

The results here are produced on `sar_sim`, an open testbed that couples the Gazebo physics simulator to an OpenAirInterface (OAI) 5G standalone radio access network (RAN) and an Open5GS core through a real-time A2G channel bridge, with each device's data path isolated in a Linux network namespace so that traffic in both directions genuinely crosses the emulated air interface (Section V). The platform is deliberately terrain-agnostic: a world is authored once and its obstacle geometry is shared between what Gazebo renders and what the channel model evaluates, so five distinct terrains can be studied under one instrument with everything else held fixed. Against that fixed platform we vary only the terrain, asking not about node altitude alone but about the full hover-versus-roam positioning decision, and we find that terrain can reverse the naive follow-the-robots intuition.

IV. Why Protocol-in-the-Loop Emulation

A. Analysis vs. Simulation vs. Emulation

Three methods are available for studying A2G links: closed-form analysis, packet-level simulation, and protocol-in-the-loop emulation with a real stack. Analysis and simulation scale but abstract away the MAC scheduler, hybrid-ARQ, link adaptation, and the RLF state machine—exactly the behaviours that turn a change in geometry into a change in user-visible throughput, latency, and outage. Because our question is precisely how those behaviours manifest as the node moves through terrain, we run the actual software and inject only the physical-layer channel, trading scale (one cell, two devices) for fidelity.

B. Why OpenAirInterface

OAI is a real, standards-compliant 5G New Radio standalone implementation, so its scheduler, hybrid-ARQ, link adaptation, and RLF handling are genuine, not modelled. Its `rfsimulator` backend replaces the radio front-end with a software channel between the base and device in-phase/quadrature streams, and, decisively, exposes a runtime channel-model interface through which per-link path loss can be set live—the hook our mobility coupling requires. Alternatives were considered: srsRAN is likewise open but its channel emulation is less amenable to the live per-link injection we need, and pure simulators lack the real protocol implementation that is the point.

C. Why Gazebo, Open5GS, and Supporting Tools

Gazebo [14] provides rigid-body physics, a publish-subscribe transport for ground-truth poses, and standard robot models. Open5GS [13] supplies a production-grade 5G core so that device attach and session establishment are genuine signalling procedures. Linux network namespaces provide the traffic-path isolation that makes bidirectional measurement honest, and `iperf3` and `ping` generate and measure the flows.

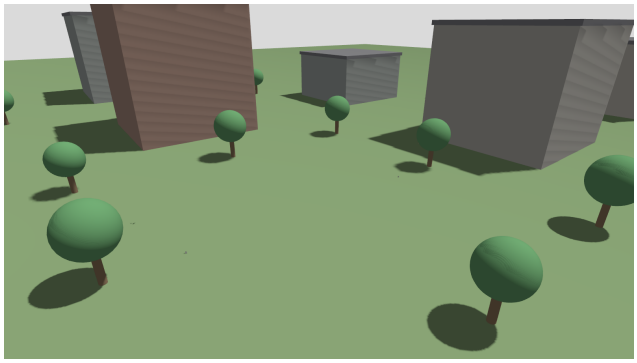


Fig. 1. A rendered terrain instance (urban). The x500 UAV carries the base station; two ground robots are the devices. The same building footprints seen here are imported by the channel bridge, so the radio shadowing matches the visible scene.

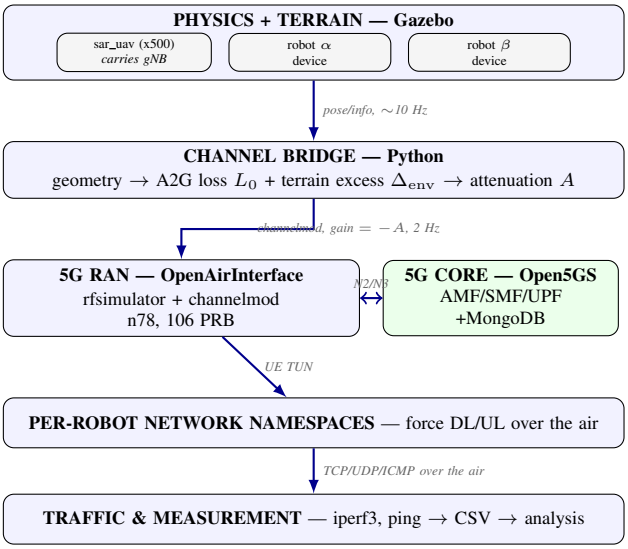


Fig. 2. Testbed architecture. The channel loop closes the coupling between physical motion in terrain and radio conditions; user traffic is measured over the emulated air interface after namespace isolation.

V. System Architecture

A. Control Loops and Data Flow

The testbed runs three concurrent control loops on one host (Fig. 2); Fig. 1 shows a rendered instance. The *channel loop* (2 Hz) reads poses, computes terrain-aware A2G path loss, and injects it into the live RAN. The *traffic loop* (about one cycle per robot every 15–20 s) exercises and measures each link. The *mobility loop* flies the UAV and drives the robots.

B. Component Communication

Gazebo publishes ground-truth poses on a transport topic at about 10 Hz; the bridge subscribes and tracks the UAV and both robots, and returns velocity commands to drive the robots. Every 0.5 s the bridge evaluates the channel (Section VI) and writes per-link attenuation to the RAN over its control interface, with a start-up read-back verifying each write. The `rfsimulator` carries the base-device sample streams and

TABLE I
SOFTWARE STACK AND ROLES

Component	Version	Role
Ubuntu Linux	22.04 LTS	Host OS (single machine)
Gazebo Sim (Harmonic)	8.x	Physics; publishes 3-D poses
gz-transport / gz-msgs	13 / 10	Pub/sub bus (Python)
PX4 Gazebo model	x500	UAV/gNB carrier
TurtleBot3 (Gazebo)	burger	Two ground-robot devices
OpenAirInterface	2026.w2x	5G NR SA RAN + rfsimulator
Open5GS	2.8	5G core (AMF/SMF/UPF)
MongoDB	6.0	Subscriber database
Python	3.10	Bridge, mobility, analysis
iproute2	5.15	Per-robot network namespaces
iperf3	3.16	TCP/UDP generation and measurement

applies the injected channel; the base connects to the core over the standard N2 and N3 interfaces, and device attach performs genuine registration and session establishment authenticated against a subscriber database. Each device terminates in a host tunnel interface that is moved into a per-robot namespace so downlink traffic genuinely crosses the radio (Section IX).

C. Software Stack and Versions

Table I lists the inventory.

VI. Geometry and the Terrain-Parameterised A2G Channel

A. Distances and Elevation Angle

With the UAV at (x_u, y_u, h_u) and a robot at (x_r, y_r, h_r), the horizontal distance, slant distance, and elevation angle are

$$d_h = \sqrt{(x_u - x_r)^2 + (y_u - y_r)^2}, \tag{1}$$

$$d_{3D} = \sqrt{d_h^2 + (h_u - h_r)^2}, \tag{2}$$

$$\theta = \arctan\big[(h_u - h_r)/d_h\big]. \tag{3}$$

Two robots at equal map distance d_h can experience very different links, because the radio responds to the slant path (2) and, decisively, the elevation angle (3).

B. Free-Space Loss and LoS Probability

The free-space loss over the slant path at carrier frequency f is

$$L_{\mathrm{FSPL}} = 20\log_{10}(d_{3D}) + 20\log_{10}(f) - 147.55, \tag{4}$$

and, following [4], the LoS probability is a sigmoid in elevation angle, giving a base loss

$$P_{\mathrm{LoS}}(\theta) = \frac{1}{1 + a\,\exp[-b(\theta - a)]}, \tag{5}$$

$$L_0 = L_{\mathrm{FSPL}} + P_{\mathrm{LoS}}\,\eta_{\mathrm{LoS}} + (1 - P_{\mathrm{LoS}})\,\eta_{\mathrm{NLoS}}, \tag{6}$$

with $a = 9.61$, $b = 0.16$, $\eta_{\mathrm{LoS}} = 1$ dB, $\eta_{\mathrm{NLoS}} = 20$ dB.

C. Terrain Excess Loss — Worked Derivations

The distinguishing element is an additive terrain term Δ_{env}, and a possible hard LoS block, evaluated on the *same* geometry Gazebo renders. Let the base-to-device segment be $\mathbf{p}(t) = \mathbf{p}_0 + t(\mathbf{p}_1 - \mathbf{p}_0)$, $t \in [0, 1]$.

a) Urban (hard blockage): Each building is an axis-aligned box $B = [x_0, x_1] \times [y_0, y_1] \times [z_0, z_1]$, inflated by a margin m to model grazing. Using the slab method, for axis i with $\delta_i = p_1^i - p_0^i$ the segment enters and exits the slab at

$$t_i^- = \frac{(l_i - p_0^i)}{\delta_i}, \quad t_i^+ = \frac{(u_i - p_0^i)}{\delta_i}, \tag{7}$$

(swapped so $t_i^- \le t_i^+$), with l_i, u_i the inflated bounds; a segment parallel to a slab is tested against its bounds directly. The box is struck iff

$$\max_i t_i^- \;\le\; \min_i t_i^+ \quad \text{within } [0, 1]. \tag{8}$$

If any building is struck, the link is NLoS: $P_{\mathrm{LoS}} \to 0$ so the full η_{NLoS} applies. Trees, by contrast, are clutter, not hard blockers.

b) Forest (Weissberger foliage): The foliage depth D is the total length of the segment inside tree canopies, each modelled as a sphere of centre $\mathbf{c}_j$ and radius r_j; the segment-sphere overlap $[t_0, t_1]$ contributes $\|\mathbf{p}_1 - \mathbf{p}_0\|\,(t_1 - t_0)$, summed over trees and capped. Weissberger's modified exponential-decay model [9] then gives, with f in GHz and D in metres,

$$\Delta_{\mathrm{env}}^{\mathrm{forest}} = \begin{cases} 0.45\, f^{0.284}\, D, & 0 < D \le 14, \\ 1.33\, f^{0.284}\, D^{0.588}, & 14 < D \le 400. \end{cases} \tag{9}$$

The link is treated as NLoS once D exceeds a small threshold (2 m), so foliage adds both graded loss and a loss of the LoS bonus.

c) Hills (knife-edge diffraction): Each hill that intrudes into the path is a single knife edge whose top stands a height h above the straight base-device line. With d_1, d_2 the distances from the two endpoints to the edge and λ the wavelength, the Fresnel-Kirchhoff parameter is

$$v = h\sqrt{\frac{2}{\lambda}\left(\frac{1}{d_1} + \frac{1}{d_2}\right)}, \tag{10}$$

and the ITU-R single-edge diffraction loss [10] is

$$J(v) = \begin{cases} 0, & v \le -0.78, \\ 6.9 + 20\log_{10}\big(\sqrt{(v - 0.1)^2 + 1} + v - 0.1\big), & v > -0.78. \end{cases} \tag{11}$$

We take $\Delta_{\mathrm{env}}^{\mathrm{hills}} = \max_j J(v_j)$ over intruding hills (worst edge dominates), capped, and treat the link as NLoS once the loss exceeds a threshold.

d) Indoor (3GPP outdoor-to-indoor): When the device lies inside the building footprint, a 3GPP-style term [8] applies a fixed wall penetration L_{tw} plus a per-metre depth loss over the distance d_{in} the device sits inside:

$$\Delta_{\mathrm{env}}^{\mathrm{indoor}} = L_{\mathrm{tw}} + \kappa\, d_{\mathrm{in}}, \tag{12}$$

with $L_{\mathrm{tw}} = 9$ dB, $\kappa = 0.5$ dB/m, capped at 40 dB, and the interior treated as deterministic NLoS. Outside the footprint the open-field model (5)–(6) applies.

D. Injection, Clipping, and Applied Gain

The total loss is $L = L_0 + \Delta_{\text{env}} + X_\sigma$, where X_σ is correlated log-normal shadow fading (larger variance under NLoS, decorrelated over distance moved). The bridge references L to $L_{\text{ref}} = L_0(50\,\text{m}, 30\,\text{m})$, adds a per-link baseline L_{base} (8 dB for α's uplink, 13 dB for the weaker β), and clips:

$$A = \text{clip}\big(L_{\text{base}} + (L - L_{\text{ref}}),\, 0,\, 45\big)\ \text{dB}. \tag{13}$$

Because the RAN applies its channel field as a linear *gain* (Section IX), the value written is $-A$, scaling each sample by $10^{-A/20}$, a true attenuation.

E. Channel Validation Against the Analytic Model

The attenuation the RAN actually applied tracks the per-sample model prediction to within about 1.2 dB mean absolute error, and independently the device-reported reference signal received power falls monotonically with injected loss—an external corroboration, since it is measured by the modem, not by us. The per-terrain model/injected/measured agreement is shown directly in Fig. 3, so the behaviour reported below is a consequence of the intended physics, not an artifact.

VII. Traffic Model and Performance Metrics

A. Traffic Types

Each cycle exercises a device with three traffic types mirroring search-and-rescue use. *Bulk upload* is a short `iperf3` TCP transfer from robot to base, modelling a sensor dump; goodput reflects reliable-transport capacity. *Video* is a fixed 8 Mbit/s uplink stream modelling a live camera; delivered rate, loss, and jitter reflect stream quality. *Command and control* (C2) is a short sequence of small pings from base to robot, emulating waypoints and emergency-stop; its round-trip time and loss measure *controllability*, not bandwidth.

B. Standard Metrics

We record device-reported reference signal received power and signal-to-interference-plus-noise ratio, uplink round-trip time, packet delivery ratio, and, per traffic type, goodput, loss, and jitter. We define *C2 availability* as the fraction of samples with less than total C2 loss, *outage probability* as the fraction in total loss, and inter-device Jain fairness over per-device mean goodput. All metrics are reported per device (α stronger, β weaker).

C. New Operational Metrics for Disaster Response

Beyond the standard set we derive four responder-facing metrics from the same logs, defined and reported in Section XI J: (i) *time-to-first-outage*, the safe window from link-up to first C2 loss; (ii) *minimum viable position*, the largest device range sustaining at least 90% C2 availability for the weaker device; (iii) *connectivity recovery time*, how long a dropped device takes to return; and (iv) a *hover-versus-roam decision rule* per terrain.

VIII. Mobility and Propagation-Aware Coverage Governance

A. Naive Geometric Safe-Radius Governor and Why It Fails in NLoS

Robots drive by differential-drive velocity commands at 1 m/s along randomised headings. To keep them from being stranded, a governor confines each robot. A naive governor inverts the open-field model for a *geometric safe radius*

$$R_{\text{safe}}(h) = \max\{\, d_h : L_{\text{base},\beta} + [L_0(d_h, h) - L_{\text{ref}}] \le A_{\text{safe}} \,\}, \tag{14}$$

with $A_{\text{safe}} = 30$ dB. This *fails* in NLoS: a robot can sit well inside R_{safe} yet be blocked by a building or canopy at loss near the cliff, because the distance rule is blind to obstacles. Once such a device declares RLF, and because a device cannot complete cell re-synchronisation while buried in deep NLoS, restarting it fails to re-attach and the recovery loops indefinitely.

B. The Propagation-Aware Governor

We therefore make the governor react to *measured* path loss. When a device's loss is elevated it is steered toward the base ground track, raising the elevation angle and reducing NLoS depth; if loss stays near the cliff the robot is relocated to a small radius from the base before it dies. Indoors, where the robot must occupy the building, it instead retreats toward the door (reducing penetration depth) and re-enters on recovery. This prevention-first design converts an unrecoverable cascade into rare, short outages.

C. Governor as a Baseline for Future AI-Assisted Control

The governor is a hand-tuned, threshold-based controller. It is deliberately simple, and it is best read as a *baseline*: the labelled corpus it produces—path loss, signal quality, block-error and hybrid-ARQ trends, and outage labels, all tied to geometry and terrain—is exactly the substrate a learned or reinforcement-learning controller would train on to place the node and route the robots. We make no learned-control claim in this paper; Section XIV sketches that direction honestly as future work.

IX. Emulation Correctness Safeguards

Building an honest single-host testbed surfaced three failure modes that produced *plausible but wrong* data; all reported results were collected only after all three were fixed.

A. The Channel Field Is a Gain, Not a Loss

The RAN's channel field is applied as a linear gain ($10^{x/20}$): positive values *amplify* until sample clipping, so a positive "loss" produced a link that degraded only through clipping and a received power that *rose* with distance. We keep A positive and negate once at injection (14).

B. Single-Host Traffic Bypasses the Radio

With host-local device addresses, base-to-robot traffic went over loopback (a "downlink" test read gigabits per second while the base scheduled a fraction of a megabit). Moving each tunnel into a per-robot namespace forces the air path in both directions.

C. Single-Client Control Port Silently Drops Writes

The RAN control port accepts one client; a second connection sat unaccepted in the backlog, silently discarding one device's channel updates and imitating a weaker device. A shared connection per port with a start-up read-back refuses to run unless every baseline verifies.

X. Experimental Setup

A. Terrain Instantiation and Shared Geometry

Five terrains are instantiated: open field, urban (building footprints), forest (tree canopies), hills (a height field), and an enterable building. Each terrain authors its obstacle geometry once; the same geometry drives both the Gazebo render and the channel excess of Section VI, so shadowing is consistent with the visible scene by construction.

B. Trial Structure

Each terrain runs a single six-hour trial in two phases: *phase 1* (0–3 h) with the node hovering statically, and *phase 2* (3–6 h) with the node performing a continuous three-dimensional random walk over a bounded box while never dwelling. Both robots free-patrol throughout, so the only variable between phases is node mobility. A single physical stack serves only one trial (two concurrent trials share the same robots and corrupt each other, so trials are serialised under an exclusive lock). Clean trials comprise 733, 721, 482, and 974 samples for open field, urban, forest, and hills; the indoor trial is reported separately as a coverage limit.

C. RAN and Core Configuration

The base operates in band n78 at 3.32 GHz, 40 MHz, 30 kHz subcarrier spacing, 106 resource blocks, connected to Open5GS over N2/N3. Exploratory runs collected while the safeguards of Section IX were latent are excluded from every number.

XI. Results

A. Propagation and Coverage Across Terrains

Fig. 3 plots path loss against base-device slant distance per terrain, overlaying the analytic A2G model, the value actually injected into the RAN, and the value the device measured from its reference-signal received power. Two things stand out. First, the injected and measured curves track one another closely in every terrain, an end-to-end validation that the device experiences the channel we intend (injected-versus-model agreement about 1.2 dB mean absolute error). Second, the terrain shapes the curve: open field and hills sit low (shallow, LoS-dominated), forest climbs steeply because a longer slant path cuts through more canopy, and urban is

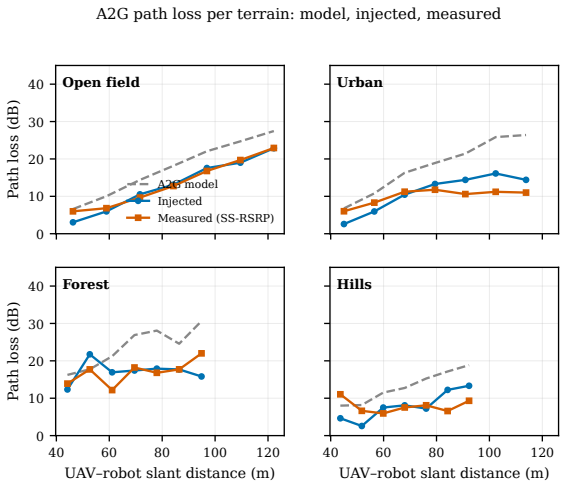


Fig. 3. A2G path loss versus base-device slant distance per terrain: the analytic model (dashed), the value injected into the RAN (blue), and the value measured by the device from reference-signal received power (orange). Injected and measured track closely; terrain sets the shape.

intermediate with building-driven jumps. Fig. 5 makes this spatial: open field is a clean radial field, urban is fractured by building shadows, forest disintegrates into thin coverage fingers, and hills form an irregular covered island. Terrain does not merely shift coverage—it changes its *shape*, so moving the node has a terrain-specific effect. Fig. 4 shows the same path loss as a time series: as the node roams (shaded), forest and hills climb toward the RLF cliff while open field and urban stay bounded.

B. Watching the Drone Move

Fig. 6 shows the raw phenomenon: both devices' C2 connectivity over the full six-hour mission, per terrain, with the static-to-roaming transition marked at 3 h (shaded = roaming). In open field the weak device β repeatedly drops while the node is static, then steadies once it roams. In urban both devices stay flat. In forest both are healthy while static, then connectivity collapses the moment the node begins roaming. Hills is a milder version.

C. Moving the Drone: The Effect Flips With Terrain

Fig. 7 and Table II give the central result: **the effect of node mobility on connectivity is terrain-dependent and changes sign with the pervasiveness of blockage.** In open field mobility *helps* the weak device: β's C2 availability rises from 56.5% (static) to 88.7% (roaming), a 32 percentage-point gain, as the roaming node drifts closer and rescues a device near the edge (mean distance falls from 91 to 75 m), while α stays near 98%. In urban it is *neutral* (about 96–98%): buildings leave LoS pockets the governor holds. In hills and forest it *hurts*, scaling with how pervasive the blockage is, -10.6 points for knife-edge terrain and -22 points for everywhere-present foliage, which collapses both devices to about 73–75%.

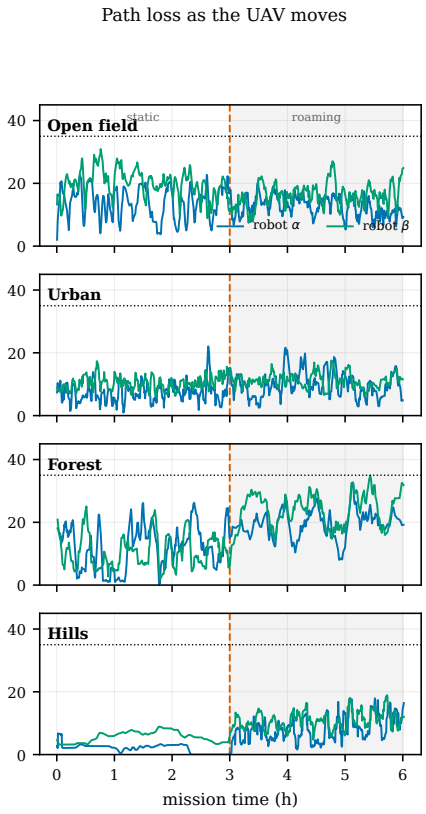


Fig. 4. Path loss over the six-hour mission as the UAV moves, per terrain, both robots; dashed line = static-to-roaming transition (3 h), shaded = roaming, dotted = RLF cliff. Forest and hills climb once the node roams.

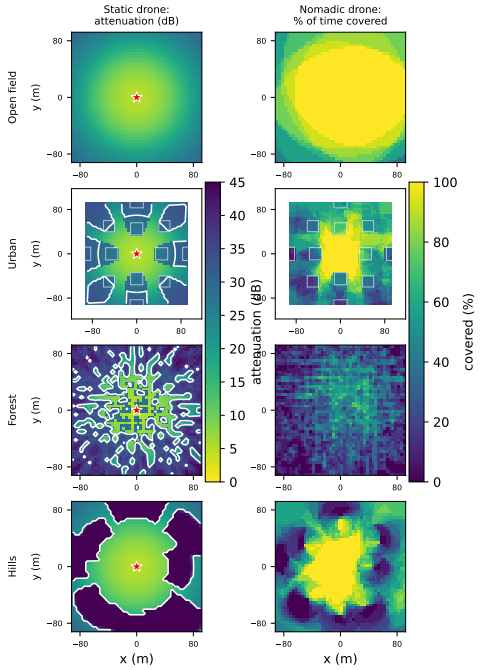


Fig. 5. Coverage field over the search area (weaker device), per terrain. Left: link attenuation for a *static* node at the origin (star; white contour = usable edge). Right: fraction of time each point is covered as a *roaming* node moves. Urban is fractured by building shadows; forest never coheres; open field spreads as the node moves.

TABLE II
C2 AVAILABILITY AND MEAN LINK METRICS BY TERRAIN AND NODE PHASE (DEVICE-AVERAGED). Δ IS ROAMING MINUS STATIC C2 (POINTS).

Terrain	Phase	C2 (%)	PLoss (dB)	SINR (dB)	TCP (Mb/s)
Open field	Static	76.8	16.5	35.1	9.7
	Roaming	93.5	14.2	38.8	13.1
	Δ	**+16.7**			
Urban	Static	97.0	8.8	43.8	17.6
	Roaming	96.9	10.4	42.6	16.1
	Δ	**−0.1**			
Hills	Static	98.9	3.9	45.7	21.6
	Roaming	88.3	9.3	44.2	19.4
	Δ	**−10.6**			
Forest	Static	96.2	10.8	40.9	15.1
	Roaming	74.2	21.7	32.5	3.2
	Δ	**−22.0**			

D. Mission Cost: Streaming and Control

Connectivity translates directly into mission capability (Fig. 8). The 8 Mbit/s video feed—the operator's eyes—arrives nearly intact in open field, urban, and hills, but in forest a roaming node drives it to 1.9 Mbit/s with over 70% loss: the operator goes blind exactly when the node moves. Command latency, by contrast, holds near 37 ms median *whenever the link is up*, in every terrain and phase, so controllability fails as an *availability* problem, not a latency creep.

E. Throughput and Coverage as the UE Moves

Fig. 9 plots TCP goodput against base-device distance, static versus roaming, per terrain. In open field the static curve falls off a coverage cliff with range while the roaming node flattens it by following the robot; in forest both curves sit low and roaming cannot lift them. Moving the node buys range where the path is clear and buys nothing where it is blocked.

F. The Real Stack Adapts

Because we run a genuine RAN, link adaptation is observed, not modelled (Fig. 10). As the forest-roaming channel degrades, the scheduler backs uplink spectral efficiency down (from 4.2 to 2.7 bits/symbol) while hybrid-ARQ holds block error near target (below 0.04%): the protocol trades rate for reliability, which a link-level abstraction would miss. This is the payoff of protocol-in-the-loop emulation.

G. Device Budget and the Weakest Radio

The two devices differ only by 5 dB of baseline budget yet experience qualitatively different networks: α is distance-limited while β is budget-limited, reaching the cliff sooner and defining the team's usable region. High Jain fairness (about

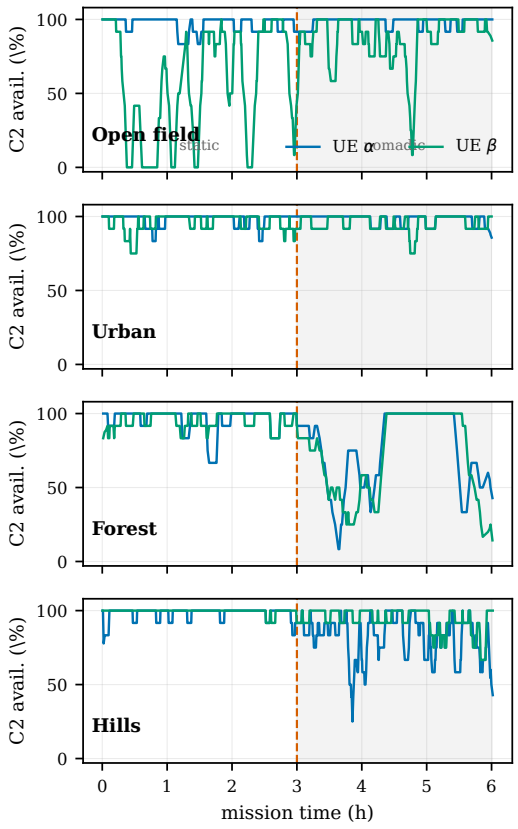


Fig. 6. C2 connectivity over the six-hour mission, per terrain, both devices (rolling availability); dashed line = static-to-roaming transition (3 h), shaded = roaming.

0.97) confirms the disparity is device budget, not scheduler bias, so the remedy is coverage margin, and node placement must be sized for the *weakest* radio, in every terrain.

H. Keeping Robots Connected in NLoS

The propagation-aware governor eliminates the RLF cascade in urban and hills (zero failed re-attaches) and contains it in forest, where the governor intervenes 574 times per trial, roughly once every 37 s, to keep robots out of foliage dead zones. The cost of prevention is a more base-centred robot distribution (the concentrated roaming coverage of Fig. 5); without it the cluttered terrains fail to produce data at all.

I. Indoor: A Coverage Limit

The building terrain is qualitatively different (Fig. 11). The fixed wall penetration plus the link baseline places the *entire* interior near the RLF cliff—depth barely matters—so shadow spikes trip RLF repeatedly, and each re-attach dominates the timeline. Lowering the node to hover directly overhead (cutting about 7 dB of A2G loss) and adding the retreat-to-door recovery removes the cascade and keeps robots alive, but C2 availability remains only 12–18%. A ground robot cannot hold connectivity deep inside a building at this link budget: leaving the building is the only recovery.

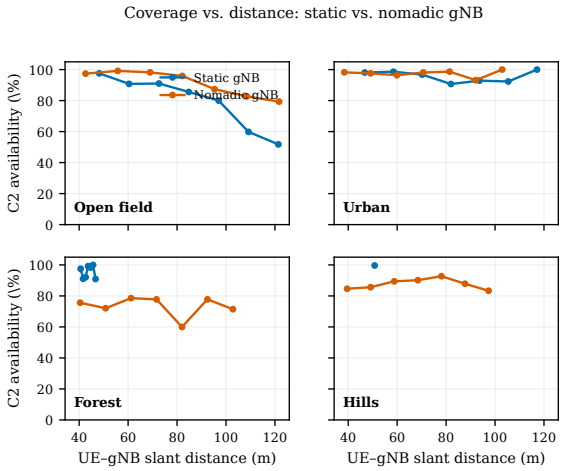


Fig. 7. C2 availability versus base-device distance, static versus roaming, per terrain. Roaming holds coverage at range in open field; urban stays high; hills degrade under roaming; forest cannot be sustained at range.

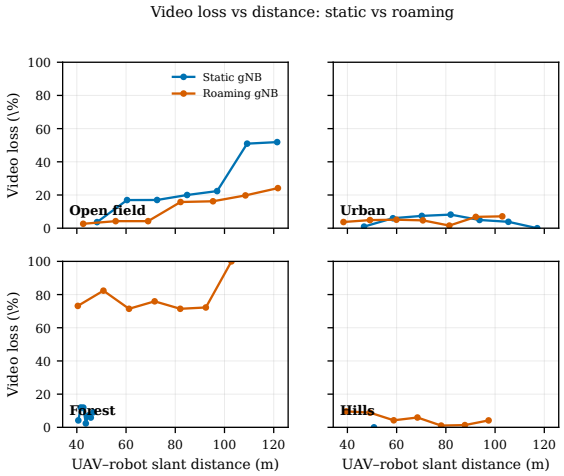


Fig. 8. Video packet loss versus base-device distance, static versus roaming, per terrain. Forest loss climbs sharply under roaming (delivered video falls to 1.9 Mbit/s); open field, urban, and hills stay low.

J. Operational Metrics for Disaster Response

We re-analyse the same logs into four responder-facing quantities (Table III).

(1) Time-to-first-outage. Measured from link-up to the first C2 loss of the weak device, this is the "safe window" after a responder flies in. It is a single-event measure and therefore noisy, but the pattern is operationally useful: in forest, a device is already at risk within minutes, whereas in open field and hills a static node buys a substantially longer window.

(2) Minimum viable position. Expressed as the largest slant range at which the weak device still holds at least 90% C2 availability, this is a deployable rule-of-thumb: keep the node within roughly 70 m of the weakest robot in open field, 90 m in urban, and 100 m in hills, whereas in forest no range sustains 90% and the node must sit almost overhead. This

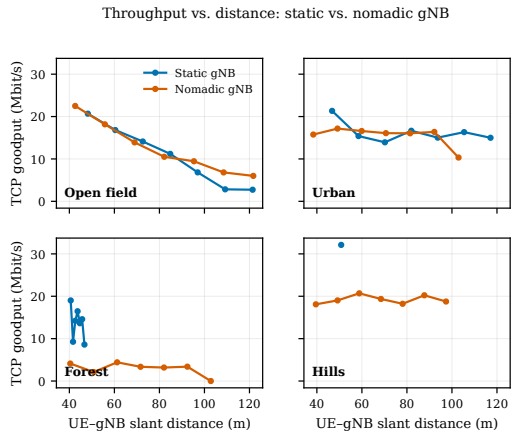


Fig. 9. TCP goodput versus base-device distance, static versus roaming node, per terrain. Roaming flattens the open-field cliff but cannot lift the blocked forest link.

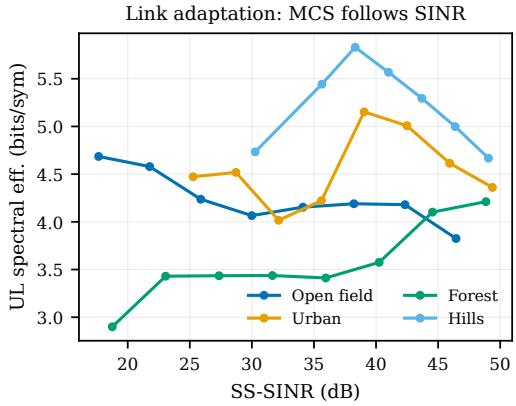


Fig. 10. Uplink spectral efficiency versus SINR per terrain. The scheduler raises the modulation-and-coding scheme with SINR; forest operates at low SINR and is forced to a lower rate, while hybrid-ARQ keeps block error bounded.

gives a responder a terrain-specific horizontal bound to hold the weakest robot inside.

(3) Connectivity recovery time. From the governor's recovery events, the mean and 95th-percentile time from detection to a restored device are 144/199 s (open field), 103/104 s (urban), 107/115 s (hills), and, worst by far, 207/383 s (forest) (Fig. 12). **Caveat.** This is a *governor-mediated* recovery time: it is floored by the 90 s sustained-loss confirmation window and the re-attach latency, the C2 channel is sampled only about every 25 s, and it therefore does not measure the RAN's intrinsic radio-link re-establishment time. Reporting the true RRC recovery time would require per-event radio-state timestamps at sub-second resolution (available in the modem logs but not in the current per-cycle CSV), i.e. a logging change or a targeted re-run; we do not approximate it from availability.

(4) Hover-versus-roam decision rule. Restating the Δ of Table II as an instruction: *open field, roam* (mobility helps the

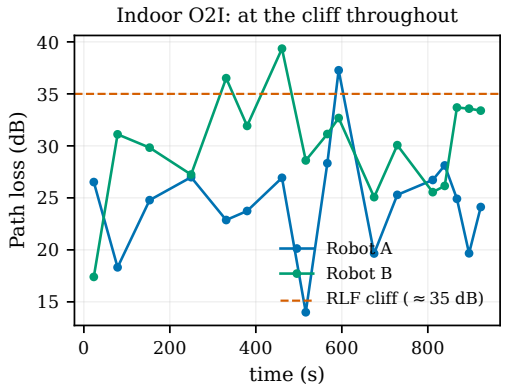


Fig. 11. Indoor penetration path loss over time (low-hovering node). Both robots sit at the RLF cliff throughout; the interior offers no reliably covered pocket.

TABLE III
OPERATIONAL METRICS FOR DISASTER RESPONSE, PER TERRAIN (WEAKER DEVICE)

Metric	Open field	Urban	Hills	Forest
Safe window, static (min)	14.8	7.7	152.7	2.7
Viable range, $\geq$90% (m)	70	90	100	<overhead
Recovery, mean/p95 (s)	144/199	103/104	107/115	207/383
Decision	roam	either	hover	hover

weak device); *urban, either* (neutral); *hills and forest, hover* (mobility hurts, most severely in forest); *indoor, hover low and overhead, and expect the deep interior to be uncoverable*.

XII. DISCUSSION

A. Why the Effect of Mobility Flips With Terrain

Moving the node changes the *propagation path*, and whether that helps depends on how blockage is distributed. In line of sight a moving node is pure geometric benefit: it can shorten links and rescue edge robots, so roaming helps the weak device most. Where blockage is *localised* (urban buildings), a governor can keep robots in line-of-sight pockets and mobility washes out. Where blockage is *pervasive* (forest canopy) there is no pocket to hold: any repositioning that lengthens the ground path lengthens the foliage path, so mobility is strictly harmful, and the damage scales with blockage density (hills below forest). Altitude still helps by raising the elevation angle, but in pervasive clutter the horizontal component of roaming can undo that gain, which is why a single follow-the-robots rule fails and the decision must be made per terrain.

B. Operating Rules for SAR Deployment

Two rules follow. First, choose hover versus roam by terrain, per Table III: roam in open ground and to rescue stranded devices, hover in foliage and rough terrain, and hover low and overhead for indoor. Second, size coverage for the weakest radio: a few decibels of device disparity relocate the usable edge and multiply the outage rate, so the node's position must protect β, not the team average.

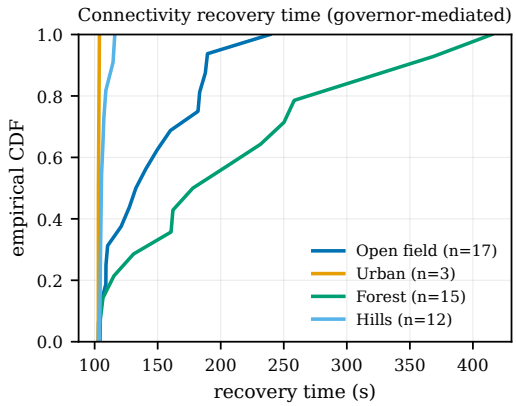


Fig. 12. Empirical CDF of governor-mediated connectivity recovery time per terrain. Forest has the heaviest tail; see the caveat in the text.

C. Implications for a Learned/AI-Assisted Controller

The terrain-dependence is precisely what makes hand-tuned control brittle and a learned controller attractive: the right node position is a function of terrain, robot geometry, and live link telemetry that our threshold governor only crudely approximates. The governor and the labelled corpus it produces are the baseline and training substrate for such a controller; we make no learned-control claim here and develop the direction as future work.

XIII. Limitations

The RAN runs over a radio-channel simulator, so absolute link budgets depend on model constants and the configured noise floor; we therefore emphasise *relative* static-versus-roaming effects within a fixed calibration rather than absolute edges. The coverage governor is necessary to obtain data in NLoS but alters robot trajectories (more base-centred), so the reported effects are conditional on prevention-based coverage keeping. We study one cell and two devices; multi-cell handover and many-robot contention are future work. Under outage the per-sample probes time out, so heavily degraded phases yield fewer samples, and the recovery-time metric is governor-mediated and coarsely sampled (Section XI J). The indoor result reflects one building geometry and link budget. Subject to these, the *direction* of the findings follows from geometry, link budget, and protocol-state mechanisms that carry to physical deployments.

XIV. Conclusion and Future Work

A. Summary of Findings

We built a mobility- and terrain-coupled 5G emulation testbed for UAV-mounted nomadic nodes serving ground robots, and used it to measure how two devices report their connectivity as the node moves, across five terrains. The effect of node mobility is terrain-dependent and changes sign with the pervasiveness of blockage: helpful in line of sight, neutral under localised urban shadowing, and increasingly harmful under diffraction and foliage, while deep indoor penetration is effectively uncoverable. We translated the result into four operational metrics and a per-terrain hover-versus-roam rule, and showed that connectivity in NLoS survives only under a propagation-aware governor.

B. Path Toward AI-Assisted Coverage Control

The clear next step is to replace the threshold governor with a learned controller that positions the node and routes robots from live telemetry and a terrain estimate, trained and evaluated on this testbed and its dataset, with the rule-based governor as its safety baseline.

C. Multi-Cell Handover, Recovery-Time Study, and SDR Validation

Three further directions are a multi-node handover study, a fine-grained radio-link recovery-time measurement using per-event radio-state timestamps (addressing the caveat above), and a software-defined-radio hardware bridge to close the simulation-to-hardware gap.

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